

Experiment 5: Cosine Similarity between Two Text Vectors

Python Code:

```
from sklearn.metrics.pairwise import cosine_similarity
import numpy as np
```

```
doc1 = np.array([[1, 2, 3, 4]])
```

```
doc2 = np.array([[2, 4, 6, 8]])
```

```
similarity = cosine_similarity(doc1, doc2)
```

```
print("Cosine Similarity:")
```

```
print(similarity)
```

```
print("\nSimilarity Score:", similarity[0][0])
```

```
if similarity[0][0] > 0.8:
```

```
    print("The two documents are highly similar.")
```

```
else:
```

```
    print("The two documents are not highly similar.")
```

Output:

```
= RESTART: C:/Users/A. MOUNESH/AppData/Local/Programs/Python/Python314/Gen
PY
Cosine Similarity:
[[1.]]

Similarity Score: 0.9999999999999998
The two documents are highly similar.
```